

How to keep infection clusters from happening

By Huang Chung-yuan and Chin Wei-chien 黃崇源 陳威全 /

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The business trip of a Briton, who contracted COVID-19 at an international conference in Singapore, turned him into a super-spreader — with an infection cluster stretching out from Singapore to France, England and Spain — and prompting the city-state to heighten its outbreak alert level in a bid to prevent the coronavirus from spreading further.

Compared with Singapore, Taiwan has not yet seen cluster or community infections, but it remains to be seen whether the nation's first asymptomatic COVID-19 case — confirmed on Feb. 9 — would make it more difficult for the government to contain the outbreak.

There is evidence that COVID-19 transmission is similar to that of influenza, which is mainly spread by droplet, contact and airborne transmission.

As COVID-19 is highly contagious, the government does not dare to overlook its ability to spread if the coronavirus strikes the nation, which is why the authorities are doing everything possible to contain the outbreak and prevent it from developing beyond any imported cases.

With the help of Google's PageRank algorithm and based on daily commute data collected in a census, our team — comprised of ourselves and National Taiwan University Department of Geography professor Wen Tzai-hung (溫在弘) — presented an epidemic risk index in an article titled "EpiRank: Modeling Bidirectional Disease Spread in Asymmetric Commuting Networks," which was published last year in the journal Scientific Reports.

Focusing on new epidemics that emerge continually across the globe, our team evaluated and ranked the infection risk of the nation's 353 townships when exposed to outbreaks.

The paper shows that the epidemic risk evaluation index is highly related to the actual spread of new epidemics over the years, such as enterovirus outbreaks, which have ravaged the nation for 20 years and posed a fatal risk to children since 1998; the 2003 SARS outbreak, which originated in China's Guangdong Province and claimed 73 lives in Taiwan; and the A(H1N1) influenza outbreak in 2009.

The results of the study demonstrate that the primary areas of infection after a new epidemic strikes Taiwan are the many administrative divisions in Taipei and New Taipei City, where mass transportation is well-developed.

With the support of scientific research and objective evidence, we, as a research team specializing in the computer simulation of epidemics, want to call on the public to fully support the government's each and every preventive measure against COVID-19.

As the outbreak has not yet developed to the stage of a cluster or community infection in Taiwan, the limited resources for disease prevention and medical treatment should be prioritized for

frontline medical personnel.

Meanwhile, the epidemic risk index for townships can serve as a reference when allocating limited medical treatment resources to potential areas of infection in case a COVID-19 outbreak occurs.

More advanced medical resources should be concentrated in the administrative areas that are more vulnerable to an epidemic, as well as the nation's political and economic centers.

In this way, the nation would be able to fully contain COVID-19 in an early stage and eliminate the possibility of it spreading.

Huang Chung-yuan is a professor at Chang Gung University's Graduate Institute of Computer Science and Information Engineering. Chin Wei-chien is a research fellow at Singapore University of Technology and Design.

Translated by Chang Ho-ming

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